

OPEN SOURCE // C++ // GERRIT

LIBREOFFICE CONTRIBUTOR FIELD GUIDE

For developers who know C++ but have never touched open source

WHO THIS IS FOR

You know C++. You've never shipped an open source patch.

This guide compresses 6 months of active LibreOffice contribution into the minimum viable path: build the codebase, find a real bug, fix it, survive Gerrit review. No filler. No motivational framing. Just the sequence that works.

THE MENTAL MODEL

LibreOffice is a C++ monorepo. Gerrit is the gate.

Every patch goes through Gerrit — a code review system at gerrit.libreoffice.org. You push a branch, reviewers comment, you iterate, it merges. The build system is **gbuild** (a Make wrapper). The codebase is ~10M LOC across modules.

LOCATION / TERM	WHAT IT IS
<code>libreoffice/core</code>	Main monorepo. Everything lives here.
<code>vcl/</code>	Visual Component Library — rendering, fonts, widgets.
<code>sc/</code>	Calc spreadsheet engine. Formula eval, row storage.
<code>sw/</code>	Writer layout engine. Text frames, pagination.
<code>cui/</code>	Common UI dialogs shared across apps.
<code>instdir/</code>	Build output. Run soffice from here.
<code>gbuild</code>	Make-based build system. Don't fight it.
<code>Gerrit</code>	Code review gate. All patches go here.
<code>tdf#NNNNNN</code>	Bug ID format. Bugzilla at bugs.documentfoundation.org .

PHASE 0 // ENV SETUP

Get the build running before touching any bug.

The build takes 45–90 min on first run. Do it once. After that, incremental builds on a single module take 2–5 min.

PHASE 0	Environment Setup	~2 hrs
01	step	Clone: <code>git clone https://github.com/LibreOffice/core ~/libreoffice</code>
02	step	Configure: <code>./autogen.sh --enable-dbgutil --enable-debug --without-java</code>
03	step	Build: <code>make -j\$(nproc)</code> — first build, go get coffee
04	step	Run: <code>./instdir/program/soffice</code>
05	tool	Set debug flags: <code>export LO_CFLAGS="-ggdb3 -O0"</code>
06	step	Configure Gerrit SSH: add key at gerrit.libreoffice.org , port 29418
07	step	Add remote: <code>git remote add gerrit ssh://<user>@gerrit.libreoffice.org:29418/core</code>
08	warn	Do NOT use <code>--enable-release-build</code> during dev — strips debug symbols

TOOLCHAIN

TOOL	PURPOSE
GDB	Primary debugger. <code>bt full/frame N/print var@10</code>
LLDB	Alternative to GDB. Prefer on macOS.
<code>clang-tidy</code>	Static analysis. Run before every Gerrit push.
<code>clang-format</code>	Formatter. LO enforces its own <code>.clang-format</code> config.
<code>valgrind</code>	<code>--callgrind</code> for CPU profiling, <code>--leak-check=full</code> for memory.
<code>heaptrack</code>	Heap allocation tracing. Faster than <code>valgrind</code> for alloc work.
<code>perf stat</code>	Linux PMU counters. Cache misses, branch mispredicts, IPC.
<code>soffice --headless</code>	Headless test runner: <code>./instdir/program/soffice --headless input.csv</code>

PHASE 1 // FIND YOUR BUG

Pick a bug you can reproduce in under 10 minutes.

Reproducibility is the filter. If you can't trigger it in a clean build, don't start. Check Bugzilla (bugs.documentfoundation.org) and filter by: **Component = VCL or Calc**, **Status = NEW**, **Keywords = bisected**.

PHASE 1	Bug Selection + Repro		~1–2 days
01	step	Search Bugzilla: filter NEW + bisected + your target module (vcl, sc, sw)	
02	step	Reproduce the bug on your build: <code>./instdir/program/soffice</code>	
03	step	Confirm the crash/regression is deterministic	
04	step	Run GDB on crash: <code>gdb ./instdir/program/soffice → run → crash → bt full</code>	
05	step	Log the full backtrace. Note the frame where logic diverges from expected.	
06	tool	<code>git log --oneline</code> — find who last touched the crash site	
07	tool	<code>git blame</code> — identify the owning developer for context	
08	warn	Avoid bugs tagged 'needsDevEval' as a first patch — high review friction	

GDB CHEAT SHEET

COMMAND	WHAT IT DOES
<code>bt full</code>	Full backtrace with locals at every frame.
<code>frame N</code>	Switch to frame N.
<code>print var@10</code>	Print 10 elements of array var.
<code>break file.cxx:N</code>	Set breakpoint at line N.
<code>watch expr</code>	Watchpoint — break when expr changes.
<code>info locals</code>	All locals in current frame.
<code>set follow-fork-mode child</code>	Follow child process on fork (LO spawns children).
<code>layout src</code>	TUI mode — source + asm side by side.

PHASE 2 // PATCH + SUBMIT

Smallest possible fix. One logical change per commit.

LibreOffice reviewers reject large patches from new contributors. Start with one function, one file. The commit message is as important as the code.

PHASE 2		Patch + Gerrit Submission	~2–3 days
01	step	Make the fix in the correct module source file	
02	step	Build only the changed module: <code>make vcl.build</code> (fast incremental)	
03	step	Run headless test: <code>./instdir/program/soffice --headless test_file</code>	
04	tool	clang-format the changed file before committing	
05	tool	clang-tidy --fix on changed file	
06	step	Commit: follow LO format exactly (see below)	
07	step	Push to Gerrit: <code>git push gerrit HEAD:refs/for/master</code>	
08	step	Respond to reviewer comments within 24h — silence = abandoned patch	
09	warn	Do NOT squash reviewer-requested changes into the original commit mid-review	

COMMIT MESSAGE FORMAT

```
tdf#NNNNNN short description of the fix Longer explanation of root cause and what
changed. Reference: https://bugs.documentfoundation.org/show_bug.cgi?id=NNNNNN
Change-Id: I... (auto-generated by commit-msg hook)
```

- Install the commit-msg hook from Gerrit — it adds Change-Id automatically.
- Subject line: lowercase, no period, imperative mood, under 72 chars.
- Body: explain WHY, not what. The diff shows what.

Gerrit review is where most new contributors quit.

The two primary reviewers for VCL/UI work are **Heiko Tietze** (UX/design) and **Jonathan Clark** (code). Heiko will challenge the UI rationale. Jonathan will challenge the implementation. Both are right more than they're wrong.

SITUATION	ACTION
Reviewer asks for refactor	Do it. Don't argue scope on patch 1.
Reviewer asks for unit test	Write it. See existing tests in the module for pattern.
Reviewer -1s the approach	Ask one clarifying question, then implement their suggestion.
No response after 2 weeks	Ping once on the Gerrit thread. Once.
Patch abandoned	Re-submit as a new patch with a fresh commit. Note prior attempt.
Change-Id conflict	Amend commit, re-push. Gerrit tracks by Change-Id not SHA.
Build fails on CI	Check Jenkins log linked in Gerrit. Fix before re-pushing.

ITERATION LOOP

```
fix → make .build → test → git commit --amend → git push gerrit HEAD:refs/for/master
```

- --amend preserves the Change-Id so Gerrit links it to the same review.

Map the codebase by abstraction layer.

PATH	WHAT LIVES THERE
<code>vcl/source/gdi/</code>	Bitmap ops, graphics context, paint path. Good first target.
<code>vcl/source/window/</code>	Window management, mnemonic assignment, event dispatch.
<code>vcl/source/font/</code>	Font loading, fallback, glyph metrics.
<code>sc/core/</code>	Calc formula interpreter, cell storage, dependency graph.
<code>sc/ui/</code>	Calc UI layer — dialogs, toolbars, sheet interaction.
<code>sw/source/core/layout/</code>	Writer page layout, frame positioning, pagination.
<code>sw/source/core/text/</code>	Writer text formatting, paragraph, line breaking.
<code>cui/source/</code>	Shared dialogs: find/replace, autocorrect, options.
<code>include/vcl/</code>	Public headers. Read before touching implementation.
<code>solenv/gbuild/</code>	Build system rules. Touch only if build is broken.

LEARNED BLOCKERS // REAL FRICTION POINTS

BLOCKER	MITIGATION
Windows native build	3-day sinkhole. Use WSL2 Ubuntu. Full stop.
gbuild + gbench	gbuild has no native benchmark support. Use out-of-tree CMake.
First build time	45–90 min. Set <code>-j\$(nproc)</code> and walk away.
Stale instdir	After a major pull, run <code>make clean</code> on the changed module.
Gerrit SSH port	Port 29418, not 22. Firewall will silently drop 22.
Debug symbols stripped	Don't use <code>--enable-release-build</code> during dev.
Mnemonic conflicts	MnemonicGenerator handles deconfliction — not <code>menu.cxx</code> .

Commands you'll use every session.

COMMAND	WHAT IT DOES
<code>make -j\$(nproc)</code>	Full build.
<code>make vcl.build</code>	Module-only incremental build.
<code>./instdir/program/soffice</code>	Run the built binary.
<code>./instdir/program/soffice --headless</code>	Headless test mode.
<code>git push gerrit HEAD:refs/for/master</code>	Push patch to Gerrit for review.
<code>git commit --amend</code>	Amend and re-push same review.
<code>git log --oneline vcl/source/</code>	Recent commits to a path.
<code>git blame vcl/source/window/foo.cxx</code>	Find author of each line.
<code>gdb ./instdir/program/soffice</code>	Attach debugger.
<code>valgrind --leak-check=full ./soffice</code>	Memory leak analysis.
<code>heaptrack ./instdir/program/soffice</code>	Heap allocation trace.
<code>perf stat ./instdir/program/soffice</code>	PMU counters: IPC, cache, branches.

LINKS

URL	PURPOSE
gerrit.libreoffice.org	Code review. All patches go here.
bugs.documentfoundation.org	Bugzilla. Find bugs, file repros.
wiki.documentfoundation.org/Development	Build instructions, module docs.
github.com/LibreOffice/core	Mirror. Use for browsing; push to Gerrit.

lxr.libreoffice.org	Cross-reference browser. Search symbols.
openhub.net/p/libreoffice	Contributor stats and activity.

Before you push to Gerrit.

PHASE —	Pre-push Checklist		every patch
01	step	Bug is reproducible on a clean build	
02	step	Fix touches the minimum necessary code	
03	step	Incremental module build passes: <code>make <module>.build</code>	
04	tool	clang-format applied to changed files	
05	tool	clang-tidy passes with no new warnings	
06	step	Headless test confirms the fix	
07	step	Commit message follows <code>tdf#NNNNNN</code> format	
08	step	Change-Id present in commit (commit-msg hook installed)	
09	step	Pushed to refs/for/master — not master directly	
10	warn	No unrelated changes in the same commit	
11	warn	No debug prints (<code>fprintf</code> , <code>SAL_WARN</code> noise) left in	

gerrit.libreoffice.org/dashboard/self | bugs.documentfoundation.org | C. SENA 2025